

VBWT Shiny App – Feature Roadmap Summary

Goal:

Transform the app into a “Where should I go birding?” decision tool that uses eBird data to guide users to the best locations based on species, time of year, and proximity.

Core Features:

- Species search (common name + 4-letter band code)
- ZIP-based hotspot recommendations with adjustable radius
- Ranking using detection rate (% of complete checklists) instead of raw sightings
- Relative frequency metric (e.g., “3× more likely than nearby average”)
- Recent activity indicator (last 7–14 days)

Timing & Seasonality:

- eBird-style frequency charts (weekly or monthly)
- Peak month indicators for each species
- Ability to compare hotspot-level seasonality to regional patterns

Habitat Intelligence:

- Assign habitat to hotspots using land cover data (e.g., forest, wetland)
- Filter results by habitat type
- Display habitat composition summaries (e.g., 60% forest, 30% wetland)

Advanced Analytics:

- Reduce bias by using complete checklists and smoothing methods
- Confidence levels based on sample size
- Identify “hidden gems” (high value, low effort sites)
- Stability indicators (consistent vs variable hotspots)
- Highlight specialty species for specific locations

Visualization:

- Interactive map:
- Marker size = checklist activity
- Marker color = likelihood
- Clustering for dense areas
- ZIP radius overlay
- Ranked hotspot table with key metrics
- Supporting charts (seasonality, habitat composition)

Species Experience:

- Display species photo (with proper attribution)
- Species profile view including:
 - photo
 - seasonal trends
 - best hotspots
 - habitat associations

VBWT Integration:

- VBWT site profiles with top species and best times to visit
- Group hotspots by VBWT sites
- “Best VBWT sites for [species]” recommendations

Outputs:

- PDF reports for sites or species
- CSV downloads of hotspot rankings
- Simple trip planning summaries

UI Strategy:

- Desktop layout:
 - sidebar (controls)
 - central map
 - details panel
- Mobile layout:
 - stacked sections (search → results → map → details)
- Responsive design for all devices

Data Strategy:

- Core dataset: eBird Basic Dataset (updated quarterly)
- Recent data: eBird API (for recent sightings)
- Habitat data: land cover datasets (e.g., NLCD)

Development Phases:

Phase 1:

- Species search
- Hotspot ranking

- ZIP search
- Seasonal charts

Phase 2:

- Relative frequency metric
- Recent activity integration
- Species photos
- UI improvements

Phase 3:

- Habitat filtering
- VBWT site profiles
- Report generation

Bottom Line:

The app becomes a smart, location-aware birding assistant that helps users quickly decide where to go and when, while supporting VBWT outreach and planning goals.